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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – Nov. / Dec. 2025

ELECTRONICS AND COMMUNICATION ENGINEERING

Fifth Semester

22LPC504 – Transmission Lines and RF Systems

Regulations 2022

(Smith chart to be Permitted)

Time : 3 Hours | Maximum Marks : 100

Answer ALL Questions

Part - A (10 x 2 = 20 Marks)

No.	Question	M	CO	BT
1.	Find the characteristic impedance of a transmission line if the following measurements have been made on the line $Z_{oc} = 550\angle -60^{\circ}\ \Omega$, $Z_{sc} = 500\angle 30^{\circ}\ \Omega$.	2	1	U
2.	What is the value of reflection coefficient for open circuit and short circuit load conditions?	2	1	U
3.	Write the condition for dissipation less line.	2	2	R
4.	State the value of attenuation and phase shift constant for the dissipationless line.	2	2	R
5.	Why double stub matching over single stub matching?	2	3	U
6.	What are the advantages of smith chart?	2	3	R
7.	Write the expression for characteristics impedance of transverse electric wave when the wave is transmitted in between two parallel plates.	2	4	U
8.	What is the cut-off wavelength of the TE_{10} mode in a rectangular waveguide?	2	4	R
9.	List the advantages of ABCD Parameters?	2	5	R
10.	Define Lossless & Reciprocal Network.	2	5	R

Part – B (5 x 16 = 80 Marks)

No.			Question	M	CO	BT
11.	a)	i)	Derive the expression for transfer impedance of a transmission line.	4	1	AN
		ii)	The primary constants for a certain transmission line operating at 7.5kHz are R = 2.6 Ω/km, L = 2.4mH/km, C = 0.0078μF/km and G = 0.11μ mho/km. At the sending end of 50km length of such a line, an ideal generator having a voltage of 10V r.m.s. is connected and the output end is terminated in a matched load. Calculate the power consumed in the load.	12	1	AN
			(OR)			
	b)	i)	A transmission line 10km long is terminated properly at the far end. At a freq. of 1000 Hz the attenuation and phase constants of the line are respectively 0.03 neper / km and 0.03 radians / km. If the far end voltage at 1000Hz is 4 Volts. Calculate the Sending End Voltage of the line.	4	1	AN

		ii)	Derive the general solution of transmission line equation can be expressed in terms of receiving	12	1	AN
12.	a)	i)	Find the sending end line impedance for a high frequency line having characteristics Impedance of 50Ω. The line is of length 1.185λ and its terminated in a load of 110 + j80 Ω	16	2	A
		ii)	A Lossless line in air having a characteristics impedance of 300 Ω is terminated in unknown impedance. The first voltage minimum is located at 15cm from the load. The Standing wave ratio is 3.3. Calculate the terminated impedance.	06	2	U
			(OR)			
	b)	i)	In a transmission line the VSWR is given by 2.5. The characteristics impedance is 50 Ω and the line is to transmit a power of 25 W. Determine the magnitude of the maximum and minimum voltage and current.	10	2	A
		ii)	A Low Loss high frequency transmission line having a characteristics impedance of 500 ohms is terminated by an impedance 600 + j 300 ohms. Determine the VSWR and Reflection Co-efficient.	06	2	U
13.	a)	i)	A lossless line of characteristic impedance 100 Ω is terminated in an unknown load impedance producing standing wave ratio of value 3. The operating frequency is 300 MHz, the length of the line is 280cm. Determine the Sending end impedance, unknown load impedance if first voltage maxima occurs at 120cm from load using Smith Chart.	12	3	A
		ii)	What is the importance of a quarter wave line?	4	3	U
			(OR)			
	b)	i)	A 75 Ω lossless transmission line is to be matched with a 100 - j80 Ω load using single stub. Calculate the stub length and its distance from the load corresponding to the frequency of 30 MHz using Smith Chart.	12	3	A
		ii)	Calculate a characteristics impedance of quarter wave transformer to match a load of 200 Ω to a source resistance of 500 Ω.	4	3	U
14.	a)	i)	Derive the expression for the field components of TE waves between parallel plates, propagating in 'Z' direction.	12	4	AN
		ii)	Mention the applications of waveguides.	4	4	U
			(OR)			
	b)	i)	A rectangular air filled copper waveguide with dimension (a x b) is (0.9 inch x 0.4 inch) cross section and 12inch length is operated at 9.2 GHz with a TE_{10} mode. Find cut-off frequency, guided wavelength, phase velocity and characteristic impedance.	12	4	AN
		ii)	Explain why TEM mode is not possible in rectangular waveguide.	4	4	U

15. a) i) Define Scattering Parameters and Derive the Scattering Matrix for Two Port Networks. 10 5 AN

ii) Write the Properties of S Matrix. 6 5 U

(OR)

b) i) Derive the Overall Network parameters for Cascade connection of two port network. 10 5 AN

ii) Why H, Y and Z parameters cannot be measured in microwave frequencies? 6 5 U

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BT	R	U	A	AN	E	C
%	12	28	22	38	-	-